

ICT energy evolution: Telecom, data centers, and AI

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Executive summary

Over the past decade, there has been considerable discussion about the electricity use of the information and communication technology (ICT) industry. There are concerns that the sector's electricity consumption might increase significantly due to increased digitalization and growing data volumes, processing, and storage. Looking back, most ICT studies have greatly overestimated future electricity use because they primarily relied on the assumption that larger data volumes would proportionally require more electricity to transmit and process the data.

Ericsson's research [\[1\]](#) shows that while the total data traffic has increased exponentially by about 80 times between 2007 and 2023, the ICT electricity consumption in the use phase has increased only 1.4 times. It is clear that data transmission and electricity use are not directly proportionate. Forecasting future ICT electricity use needs to rely on an understanding of hardware market statistics together with actual company-reported electricity consumption figures to more reliably identify existing and future energy trends.

It is anticipated that electricity use of networks and data centers will continue to increase during the coming years. However, Ericsson's analyses do not predict an exponential rise in electricity consumption as a result of increased digitalization, network expansions, and the growth of traditional data center services. The uncertainty lies in how AI services and technologies will evolve in the coming years. In the short term, there is a limitation in AI hardware supply to enable an exponential increase in energy use. It is also anticipated that more efficient AI models and hardware will be required to meet profitability, cost, and other prerequisites. Therefore, the role of technological advancements in the digital sector, coupled with well-grounded projections, is crucial for understanding the future and enabling informed policymaking to support sustainable digital growth.

Introduction

The ongoing digitalization has spurred concerns about the ICT sector's use of electricity and the corresponding emissions of greenhouse gases. Projections for future electricity demands have been developed over the past decades. However, what most of them have in common, is that they have failed to correctly predict the electricity consumption trend of the ICT sector, and the results have often been exaggerated.

This white paper will show why predictive studies often fail, how the actual electricity use of networks and data centers has evolved during the past decade, and describe what drives the electricity consumption in these technologies. We will show that data volumes and electricity use do not directly correlate, and explore potential future scenarios for the industry, focusing on networks and data centers. Furthermore, the paper will attempt to explore the impact of AI, and what can be said about its current and future energy use.

Historic and current electricity use

The ICT sector consists of three main parts: fixed and mobile networks, data centers, and user devices, such as smartphones and computers. Ericsson's research shows that the sector has continuously increased its use of electricity by approximately one to two percent per year since 2007. At the same time, there have been constant improvements in the energy performance of products and solutions. While total data traffic has increased exponentially, by approximately 80 times between 2007 and 2023, global ICT electricity consumption has increased only 1.4 times. During the same period, the number of users and the number of devices they use have increased from about 3 billion fixed and mobile subscribers to more than 10 billion. Hence, the electricity consumption per subscriber has decreased significantly. The related ICT sector greenhouse gas (GHG) emissions have followed the same trend as its energy consumption. However, since 2020, the total estimated GHG emissions for the entire lifecycle have started to decrease. The ICT sector is energy- and carbon-lean, consuming only approximately 4 percent of the world's electricity and contributing to 1.4 percent of greenhouse gas emissions. [\[2\]](#)

ICT sector electricity consumption 2020 - 2023

In 2023, the entire ICT sector's electricity consumption in the use phase was approximately 1,000 TWh. [\[3\]](#) This is a slight increase from 2020 when approximately 940 TWh were reported. Networks and data centers contributed approximately 550 TWh to the total electricity usage in 2023, and the remaining 450 TWh was from user devices. These results are based on research performed by Ericsson, combined with publicly disclosed electricity use reports from more than 160 large ICT companies, representing more than 90 percent of the internet traffic. To achieve a global representation, the disclosed figures have been extrapolated to create a total for each subsector.

Collecting and analyzing disclosed information from network operators and data center operators provides an opportunity to explore the development of these segments. Figure 3 shows reported numbers from 63 network operators, together covering about 75 percent of all mobile and fixed subscriptions globally. The figures show a small increase in electricity for 2023 compared to 2020. The right graph in Figure 3 shows the electricity consumption numbers collected from 36 large data center operators, covering more than 90 percent of all internet traffic. Here, the increase in electricity consumption is significant, with approximately 50 percent.

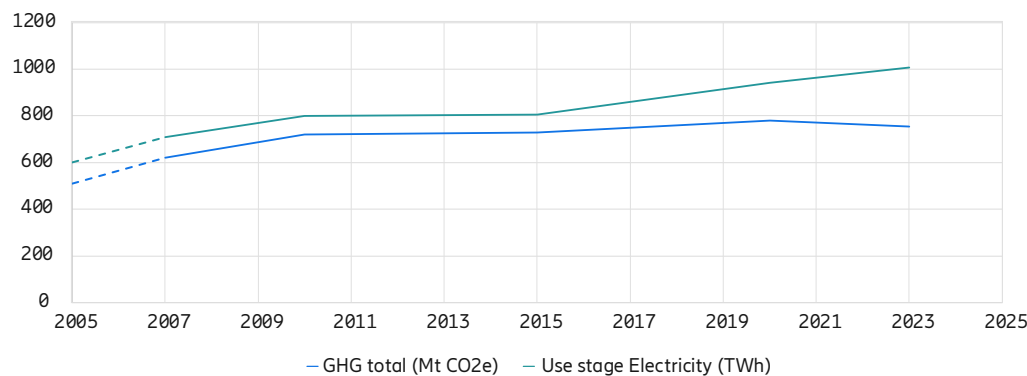


Figure 1. ICT sector development from 2007 to 2023 (pre-2007 figures are estimates)

According to Omdia Research, in 2023, a handful of companies, such as Microsoft, Meta, Google, and Oracle, procured most of all new AI GPUs. It is estimated [4] that the electricity consumption dedicated to AI-specific services, in 2023 was approximately 8 percent of the total electricity consumption of data centers, meaning less than one percent of the total ICT sector.

Total electricity consumption (TWh) of the ICT sector (including measured and reported, and estimated and extrapolated figures)

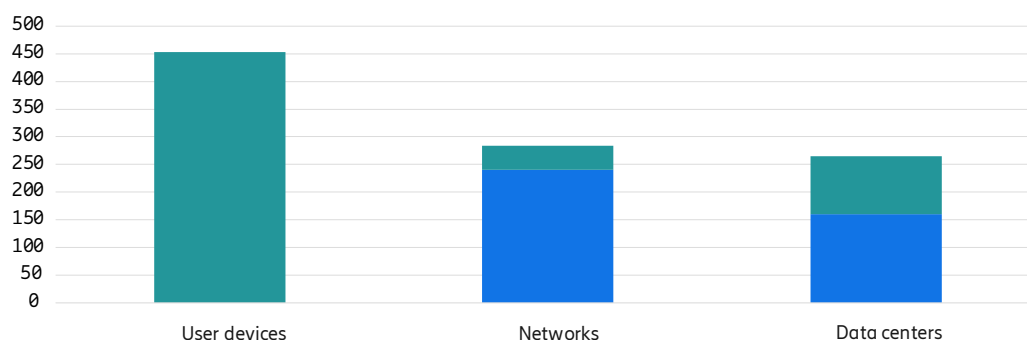


Figure 2. Total ICT sector use stage electricity (including the use stage of devices) for 2023.

Historically, there have been several studies developing projections for the electricity use and carbon footprint of the ICT sector for different end-years. In 2024, Ericsson published a study on the actual outcomes [5], showing that many projections for 2020 had overestimated the electricity consumption and greenhouse gas emission growth of the sector. The main reason for these overestimations tends to be the quality of data used in the research. Many studies primarily rely on estimated future data volumes, with the assumption that larger data volumes will proportionally require more electricity to transmit and process the data. Such methodologies lead to wrong conclusions, as will be described in section 4.

Measured and reported electricity consumption from 63 network operators and 36 data center operators

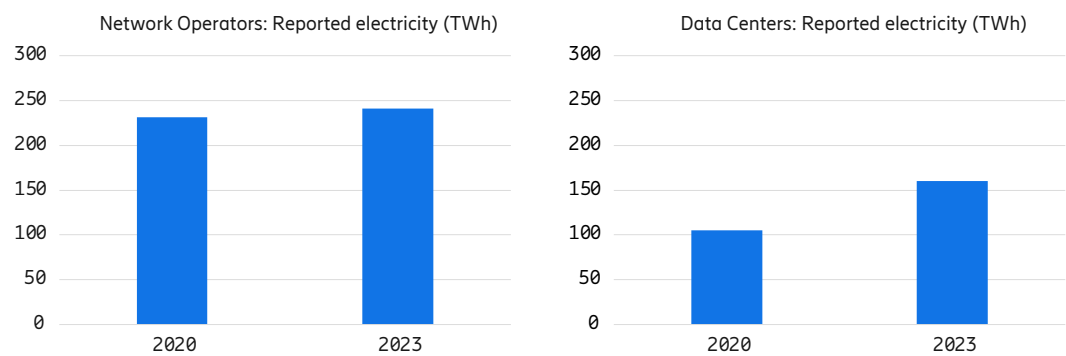
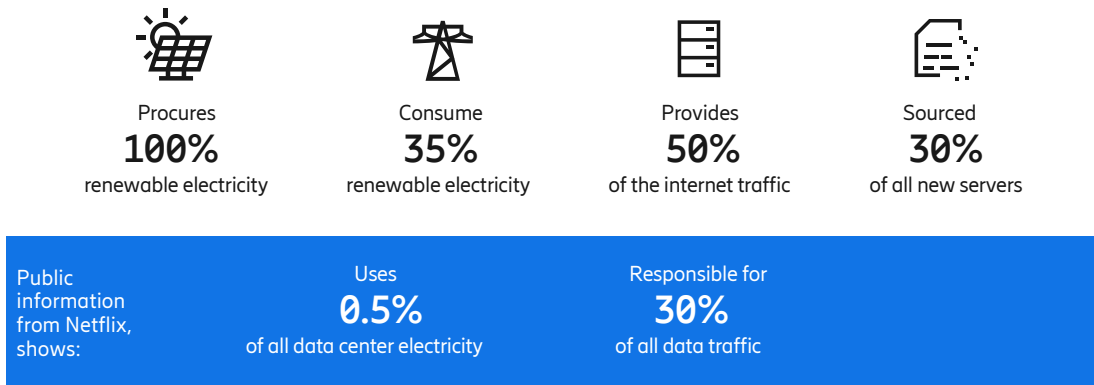


Figure 3. Electricity consumption based on publicly available and reported numbers from network operators and data center providers (crypto mining excluded) globally

Insights gathered in the study from analyzing disclosed information

The five of the largest ICT companies (Microsoft, Amazon, Google, Meta & Apple):



Why is energy use not growing exponentially?

The simple answer is that energy use is not directly proportional to data volumes. In networks, transmitting and processing data only requires a smaller part of the electricity consumed. Most of the electricity is used to provide coverage, even when no data is being transmitted or processed, this is referred to as baseload consumption. For example, in mobile networks, the cells communicate with devices, such as mobile phones, to synchronize position, signal strength, and other parameters, with only small amounts of signaling data being exchanged. This baseload consumption represents approximately 95 percent of the electricity use, where the network is monitoring all devices connected to the individual cells in the network.

Predictions of future electricity consumption using top-down studies, typically use baseline energy consumption estimates related to an intensity figure such as energy per data unit (kWh/GB). As a next step, it is assumed that current energy intensities will remain constant in the future. This is used to project future energy consumption based on data volume growth, with various adjustments for energy efficiency gains. Most of these past studies have substantially exaggerated the energy consumption of the industry. Hence, top-down approaches are not a good representation of future network energy use and cannot be used to calculate the energy consumption of real-world applications.

An alternative approach is to utilize bottom-up studies, using detailed data on technology, such as equipment specifications of server power use, data center infrastructure characteristics (such as power usage effectiveness – PUE), and installed base and equipment shipment values. They tend to be closer to the actual outcome from networks

and data centers. However, it is important to combine the bottom-up approach with a reality check, utilizing reported numbers from companies to verify if the model is a good representation of real systems.

Energy efficiency and cellular communication standards

The electricity consumption in cellular networks is largely dependent on the generation of the telecommunications standard. For example, for 4G, system signaling is transmitted every 0.2 milliseconds (ms), to communicate with any devices. On the other hand, in the 5G standard, the time between mandatory transmissions can be 20ms or even longer – 100 to 800 times longer than 4G. The extended communication intervals have resulted in the opportunity to utilize sleep modes for the radio access network, resulting in a corresponding gain in reduced electricity consumption. For the future 6G, there is an opportunity to enhance these communication intervals even more, and thus achieve further energy efficiency gains. [6]

The bottleneck is that users require fast access to communication networks. Hence, the intervals cannot be so long that they interfere with the user experience, such as being able to use a service or make a phone call when needed.

Estimating the electricity consumption

Estimation of electricity consumption of networks and data centers is recommended to be based on a two-step approach, combining a bottom-up analysis with actual reported energy consumption figures. The bottom-up analysis, also called the power model, has been described in peer-reviewed articles.[7] In principle, the power model utilizes information on individual hardware components, considering their maximum power and idle mode power values to calculate a representative energy use over time across the network or in a data center. The second step is a reality check, where current company-reported figures are used to understand if the values obtained through the power model are correct. Assessing company-reported data can be supplemented with country-wide statistics on energy production and consumption. For example, is an estimated growth of the sector for a certain historic time period reflected in public statistics? The reality check is important to obtain values that correspond to actual conditions, as well as to understand several parameters, such as the utilization ratio in data centers, and to assess the overall projection model used.

Furthermore, an important energy efficiency indicator to observe in the long term is the energy consumption per subscriber (kWh/subscriber), since the number of subscribers does not significantly vary over time. As described previously, most of the electricity in a network is used to keep the system up and running, without any data being transmitted. For example, GSMA reported during the Covid-19 pandemic, that data transmission in mobile networks increased by 50 percent, while electricity consumption remained flat.[8] In data centers, it is important to assess the actual power capacity used, which is often much lower than its maximum capacity. Additionally, the main difference, from an energy perspective, between a telecommunications network and a data center is the possibility to scale up and down the capacity, and related electricity use.

Another aspect that is not fully considered in many future predictions is the constant development and efficiency improvements of ICT equipment. Network equipment and new technology generations are becoming more energy-efficient and increasing their capacity

and computational power, requiring less equipment to meet performance requirements. The same applies to servers, routers, switches, and other infrastructure. This is an important factor determining the overall electricity consumption for future infrastructure.

In conclusion, even if we increase data transmission volumes exponentially, we will not see the same increase in electricity consumption. This has been true historically and will likely be true in the future.

What about the future?

Future projections are complicated, and several factors need to be considered, such as the expansion of network coverage and data centers, the type of equipment used, the number of subscribers, and the technological evolution of hardware and system generations. The power model, combined with real data, as discussed in previous sections, is currently the most accurate approach available. Similarly, for networks, it has been shown that using intensity values (such as energy per transmitted data) will lead to overestimations and incorrect outcomes.

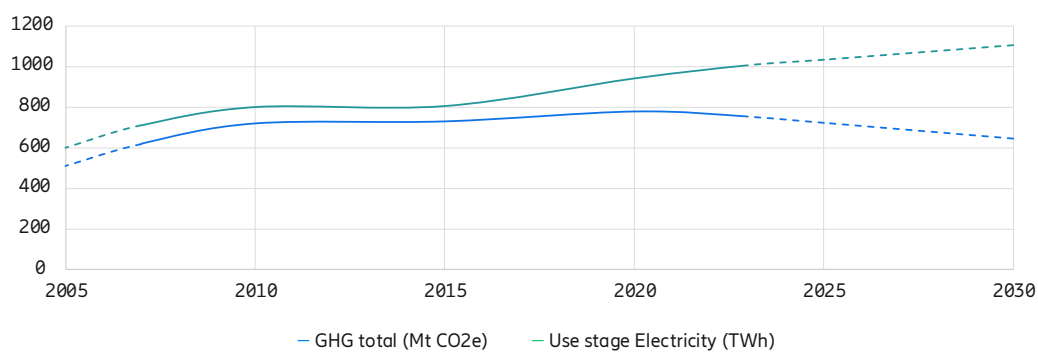


Figure 4. The ICT sector use stage electricity consumption and total carbon footprint, without the potential increase from AI, with forecast until 2030

The increased digitalization and the need for more network coverage, the transfer of storage and compute from local to cloud services, and an overall higher demand for high-quality data services will increase the electricity consumption by digital services. Ericsson has estimated the electricity use and potential carbon footprint for the ICT sector, up to 2030, see Figure 6. It is expected that the overall electricity consumption will continue to increase by approximately 10 percent, from 2023 until 2030, to approximately 1,100 TWh. For networks and traditional data center services, an increase in electricity consumption of 6 percent and 13 percent, respectively, is anticipated until 2030.

In addition, there is an expansion of services utilizing AI, and we have only seen the beginning of this evolution. However, for networks and traditional data center services, even if we expect the sector's electricity consumption to increase, it is not anticipated to be exponential, as some studies have reported. The simple reason is that equipment is constantly becoming more efficient, standards for new generations of network systems are increasingly incorporating energy efficiency measures, and lastly, data transmission will never directly correlate with electricity use. Furthermore, as we build and deploy new generations of networks and server systems, older and less energy-efficient technology generations and hardware will be phased out and replaced with more energy-efficient equipment.

And the impact of AI?

How will the uptake of AI impact the data center market in the coming years? First, the technology is still in its infancy and developing rapidly, making it difficult to predict how AI systems will be built in the future. Current systems use large amounts of electricity for training purposes and utilize continuously more powerful hardware. However, we have also seen potential disruptions, such as DeepSeek, which claims it can run large language models (LLMs) much more efficiently than other players. Furthermore, huge amounts of investments have been announced in the development of AI, which will also be used to make the technology more efficient. From a business perspective, it is a necessity to lower the energy costs of AI to be able to have future attractive offerings and sound business models.

Ericsson is estimating that up to 12 million AI graphics processing units (GPUs) were in operation at the end of 2023, consuming approximately 21 TWh, corresponding to approximately 8 percent of the electricity consumption of all data centers, meaning less than one percent of the total ICT sector. Looking forward, it is estimated [\[9\]](#) that the AI energy consumption will increase to 20 percent in 2028 of the total electricity use of data centers.

In 2024, high-end GPU sales are estimated to have tripled from 2023 levels, reaching around 2 million units and matching the sales of low-end GPUs.[\[10\]](#) It is expected that high-end GPU sales will continue to rise in the coming years. 70 percent of these highly energy-consuming units are currently procured by a few large global companies. At the same time, we have seen a decline in the traditional server systems market.

Furthermore, another important aspect is the frequency and duration of AI system usage. The most energy-intensive activity is training an AI model; however, in the future AI inference or usage, is expected to be the larger part of the energy consumed.

Finally, it is important to assess the reliability of predictions of AI electricity use. Based on bottom-up analyses, scrutinizing market data for server and GPU sales, combined with ICT companies' reported energy consumption figures as well as country-wide electricity consumption statistics, we can gain a better understanding of what is plausible for the future. Since a large part of the operational costs of data centers are related to electricity, older and less energy-efficient hardware needs to be replaced to reduce operational costs and increase competitiveness. Additionally, the technological advancement needs to be accounted for. According to the International Energy Agency (IEA), the efficiency of AI-related chips has doubled every three years, and recent chips use 99 percent less power to perform the same calculation as a model from 2008. [\[11\]](#) Hence, it is highly probable that both servers and AI GPUs will advance with more efficient processors entering the market before 2030.

Conclusion

With an increased rate of digitalization, the ICT sector, particularly networks and data centers, is likely to see an increase in electricity use. Still, this increase will not be as significant as some reports have claimed. The uncertainty for the future is how AI services and technologies will evolve in the coming years.

Today, the entire sector uses about 4 percent of the world's electricity. It is estimated that electricity consumption of networks and traditional data center services will grow by one to three percent each year until 2030. To manage operational costs, older equipment will continuously be upgraded to newer, more energy-efficient equipment. AI technologies will also be used in networks and data centers to increase efficiencies in all aspects, including reducing energy consumption. Legacy technologies with low energy efficiencies, such as fixed telephony, 2G, and 3G will be phased out in favor of energy-efficient standards such as 5G. This evolution will reduce electricity use and at the same time increase capacity in networks and data centers.

Estimation of future electricity consumption trends within the ICT sector is a difficult task where several aspects need to be considered. Historically, many forecasts have overestimated electricity consumption by relying on inferior methodologies, for example using intensity figures or not comparing results with the actual company reported numbers.

Methodologies have also failed to consider how technologies function, technological advancements, and future service needs.

The adoption of AI can be a game changer for digital technologies and services. Due to the computational power required for the training of LLMs and eventually AI inference, electricity use is a concern. There will be a need to develop more efficient AI hardware and models, to reduce energy costs and improve the business case for AI services. It is important to distinguish between dedicated AI models serving a specific purpose, with LLMs. Dedicated AI services with the purpose of, for example, managing energy use in telecom networks or creating efficiencies in power grids have significant potential for creating sustainable efficiencies.

Currently, a few large companies, such as Microsoft, Amazon, Google, and Meta - representing 70 percent of all internet traffic - are procuring 80 percent of all high-end AI

hardware. At present, only a few companies are offering such hardware, causing a limitation in the supply in the short term. Based on market statistics and how AI models are operated, we expect an increase in the electricity use of AI services for years to come. Given that AI is still in an early stage, with continuous energy efficiency improvements in AI hardware and software, it is difficult to predict how big this increase will be, and how the energy consumption will evolve in the future. However, history tells us that future predictions of the evolution of hyped technologies can be overestimated, such was the case at the beginning of the 2000s during the dot-com bubble.

Decisions on investments in digitalization and future regulatory policies depend on accurate information and assumptions. The electricity use of the ICT sector is an aspect that needs to be considered, which is why reliable predictions are an important basis for decision-making. Connectivity and digital solutions that support utilities, power grids, transport systems, buildings, and industries will be vital for the green transition. Digitalization needs to be prioritized and incentivized to help phase out fossil fuels and decarbonize our societies.

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Authors



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